

REACT JS (Assignment 3)

Aim

To develop a React application named **Score Calculator App** using React Components and Props to display a student's details and calculate the percentage based on the total marks obtained.

Procedure

1. Installed Node.js and npm.
2. Created a new React application using:
npx create-react-app scorecalculatorapp
3. Opened the project in Visual Studio Code.
4. Created a folder named **Components** inside the **src** folder.
5. Created a folder named **Stylesheets** inside the **src** folder.
6. Created the component file **CalculateScore.js**.
7. Created the stylesheet **mystyle.css**.
8. Modified **App.js** to render the CalculateScore component.
9. Passed student details through props.
10. Calculated percentage and displayed the output.
11. Executed the application using:
npm start
12. Verified the output in the browser.

```
PS C:\Users\Lenovo\scorecalculatorapp> npm start

> scorecalculatorapp@0.1.0 start
> react-scripts start

(node:25124) [DEP_WEBPACK_DEV_SERVER_ON_AFTER_SETUP_MIDDLEWARE] DeprecationWarning: 'onAfterSetupMiddleware' option is deprecated. Please use the 'setupMiddlewares' option.
(node:25124) [DEP_WEBPACK_DEV_SERVER_ON_BEFORE_SETUP_MIDDLEWARE] DeprecationWarning: 'onBeforeSetupMiddleware' option is deprecated. Please use the 'setupMiddlewares' option.
Starting the development server...
Compiled successfully!

You can now view scorecalculatorapp in the browser.

  Local:            http://localhost:3000
  On Your Network:  http://192.168.0.103:3000

Note that the development build is not optimized.
To create a production build, use npm run build.

webpack compiled successfully
PS C:\Users\Lenovo\scorecalculatorapp>
```

```
▼ SCORECALCULATORAPP
  > node_modules
  > public
  ▼ src
    ▼ Components
      JS CalculateScore.js
    ▼ Stylesheets
      # mystyle.css
      # App.css
  JS App.js
  JS App.test.js
  # index.css
  JS index.js
  logo.svg
  JS reportWebVitals.js
  JS setupTests.js
  .gitignore
  {} package-lock.json
  {} package.json
  ⓘ README.md
```

Files Created

1. **App.js**
import './App.css';
import CalculateScore from './Components/CalculateScore';

```
function App() {

  return (
    <div className="App">
      <CalculateScore
        Name="John"
        School="ABC School"
        Total={450}
        goal={5}
      />
    </div>
  );
}
export default App;
```

2. Components/CalculateScore.js

```
import React from "react";
import "../Stylesheets/mystyle.css";
function CalculateScore(props) {
  const average = props.Total / props.goal;
  return (
    <div className="box">
      <h1>Student Details</h1>
      <h3>Name : {props.Name}</h3>
      <h3>School : {props.School}</h3>
      <h3>Total Score : {props.Total}</h3>
      <h3>Number of Subjects : {props.goal}</h3>
      <h2>Average Score : {average}</h2>
    </div>
  );
}
export default CalculateScore;
```

3. Stylesheets/mystyle.css

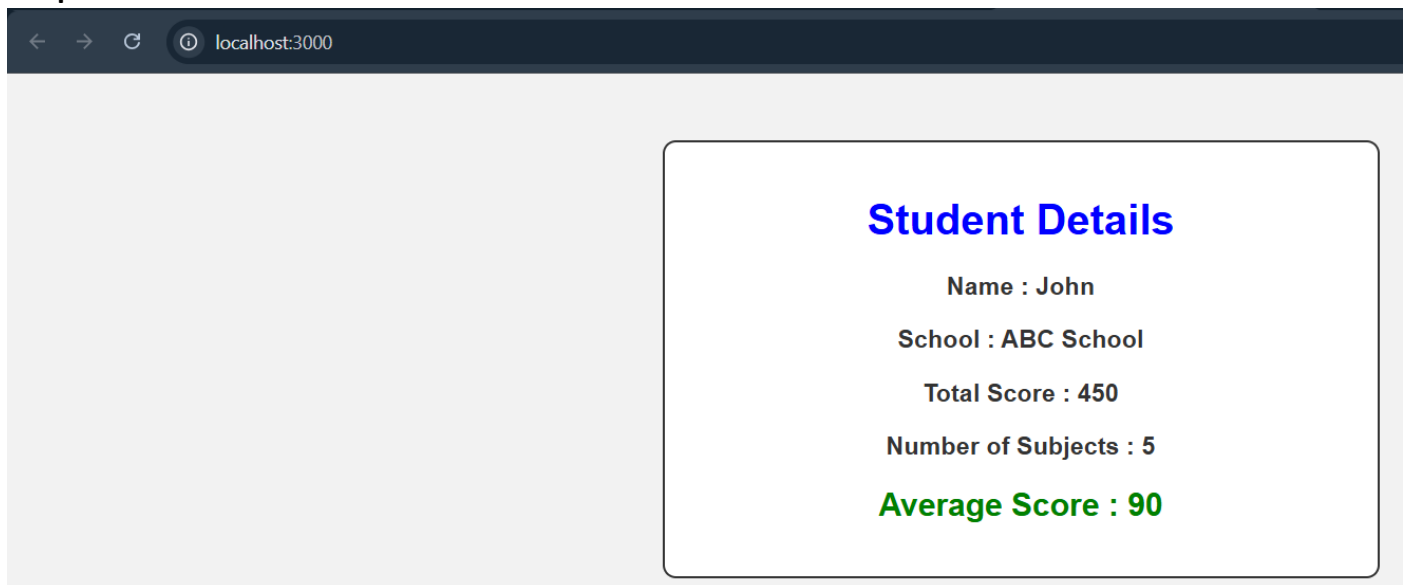
```
body {
  background-color: #f2f2f2;
  font-family: Arial, Helvetica, sans-serif;
}
.box {
  width: 500px;
  margin: 50px auto;
  padding: 20px;
  border: 2px solid #333;
  border-radius: 10px;
  background-color: white;
  text-align: center;
}
```

```
h1 {  
  color: blue;  
}  
h2 {  
  color: green;  
}  
h3 {  
  color: #333;  
}
```

Result

The React application was developed successfully. The student details and calculated percentage were displayed correctly using React components and props.

Output



Conclusion

The experiment was completed successfully. A React application was created using reusable components, props, and CSS styling to display the student's information and percentage.